



MIDNIGHT HACK BUENOS AIRES

VELO

Zero-knowledge forensic attestation on Midnight

[Contract live on Midnight preview](#)

**The verdict is visible.
The victim is not.**

2026 · v0.1.0

THE SITUATION

A forensic expert opens a case.

Abuse material. A leak. Fraud.

She runs her analysis and reaches a finding.

Now she has to prove that finding to a court.

CASE FILE



*Everything inside is private,
and most of it has to stay that way.*

AND SHE FACES A TERRIBLE TRADE-OFF

Disclose the raw evidence

So everyone can verify her work — including private messages, photographs, and information that never needed to be exposed in the first place.

Keep the evidence private

And ask everyone to trust her process. The court gets a conclusion it cannot check, from a person it has to take at their word.

VELO chooses neither.

THIS ISN'T ONLY A FORENSICS PROBLEM

How do you prove that a rule was followed without disclosing the private data that the rule was designed to protect?

Medical records

Whistleblower reports

Compliance

Any workflow where the data
is sensitive

We chose forensics because it is the sharpest version of that problem — and because it is a domain we understand deeply.

THE MECHANISM

The evidence never moves.

01**Local analysis**

A deterministic engine runs on the expert's own machine. The raw evidence never leaves it.

**02****Sealed**

The analysis is sealed into a commitment, bound to a hash-chained record of custody.

**03****Midnight**

A commitment and a zero-knowledge proof reach the network.

What reaches the network is a commitment and a proof. Never the evidence.

VELO doesn't just prove
when the rules are satisfied.

**It refuses to prove
when they're not.**

REFUSAL ONE — CORROBORATION

Same evidence. A score high enough to produce MALICE.
But there is only one independent source.

WHAT THE SCORE ALONE WOULD SAY

MALICE



WHAT VELO ATTESTS

SUSPICION

Not a warning. A refusal to produce the proof.

REFUSAL TWO — PROVENANCE

The same evidence. The same score.

This time, there is no valid chain of custody.

WHAT THE SCORE ALONE WOULD SAY

MALICE



WHAT VELO ATTESTS

ABSTAIN

The system does not say “be careful.” It refuses to make the stronger claim, because the provenance requirements were not satisfied.

AND THAT RULE ISN'T DOCUMENTATION

Two independent sources for a serious finding — enforced inside the ZK circuit.

```
contracts/velo.compact
```

```
assert(verdict != MALICE || corroborationCount >= 2)
```

The system cannot produce a valid MALICE attestation unless that constraint is satisfied.

That is not a promise from us. It is a constraint enforced by the proof system.

AND THIS ISN'T A DIAGRAM EITHER

The contract is deployed on Midnight's preview network.

CONTRACT ADDRESS

46cac58c4eb0e034b4211d754bfe67f7e8e1aa08d448ebd089437ed573023d9d

Compiled with compact 0.31.1

Real prover and verifier keys

Seal → attest → verify, end to end

WE ATTACKED OUR OWN IMPLEMENTATION

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ADVERSARIAL AUDIT ROUNDS

Path traversal

Read and write, through the case identifier. Exploited end to end over the real protocol, not theorized.

Independence that wasn't

It was counting detector categories, not sources. One artifact reached MALICE with “corroboration 4.”

Silent counter inflation

A re-attestation path let the same finding inflate a public counter, manufacturing the look of corroboration.

We fixed them, and documented what remains open in the repository.

We don't hide the failure modes. We make them part of the audit trail.

Nobody buys a zero-knowledge proof.

A forensic laboratory, a prosecutor's office
or an insurer buys verifiable trust.

THE PRODUCT

Think of VELO as a professional wallet.

A crypto wallet

The asset is money.

It never sends your private key across the network. It sends a proof that you are authorized to act.

VELO

The asset is a sealed, verifiable case.

It never sends the evidence. It sends a proof that the finding was produced under the required rules.

QUERYING A PROFESSIONAL'S RECORD WITHOUT SEEING THE CASES

How many findings were accepted?

Under which jurisdiction?

Which audits verified the process?

Was an evidence package altered after it was sealed?

None of those questions requires exposing the underlying photographs, names or private messages.

These are examples of what that interface could report — not metrics we are claiming today. What is real today is the mechanism that makes those claims possible without disclosure.

“HOW DO YOU KNOW THE PERSON ISSUING THE FINDING IS ACCREDITED?”

Not facial recognition

A face proves that someone has a face. It does not prove that they are authorized to issue a forensic finding.

A credential

A zero-knowledge proof of membership in an accredited-experts registry. It proves the issuer is authorized without revealing who that person is.

The expert gets to do her job.

The victim doesn't have to become part of the evidence
just so the process can be trusted.

The verdict is visible.
The victim is not.

